EXCEL MINOR PROJECT REPORT

Sales Operations Analytics

1. Abstract

This project uses Microsoft Excel to analyse sales operations data for a multi-category retailer operating across six Indian cities. The workbook combines order-level transactions with customer, product, sales-representative, and monthly target information. It focuses on Excel lookup functions, data-quality diagnostics, conditional aggregation, PivotTables, and advanced analysis such as ranking and running totals. The purpose is to turn raw, imperfect business data into useful information for sales and operational decision-making.

2. Problem Statement

Retail sales data can contain duplicate order IDs, inconsistent city spellings or letter case, customer names with extra spaces, invalid references, discontinued products, and transactions with unusual discount patterns. These issues can affect reports and business decisions. The project applies Excel tools to connect multiple sheets, check data quality, calculate revenue measures, compare performance with targets, and answer business questions.

3. Objectives

- Retrieve related information across worksheets using VLOOKUP and INDEX-MATCH.
- Identify duplicate order IDs, inconsistent text formatting, missing references, and suspicious order patterns.
- Calculate revenue and order counts using SUMIFS, COUNTIFS, AVERAGEIFS, and conditional IF logic.
- Summarise sales by region, product category, month, and quarter using PivotTables and equivalent formulas.
- Rank sales representatives, calculate cumulative revenue, and answer multi-condition top-N questions.
- Present clear findings while recognising that data-quality flags require review before conclusions are drawn.

4. Dataset Description

The workbook contains five main data sheets and five question sheets, in addition to a README sheet.

Worksheet	Approximate records	Purpose / key fields
Orders	2,000 orders	Order ID/date, customer, product, sales rep, quantity, price, discount, final amount, payment status, shipping city, and helper fields.
Customers	300 customers	Customer ID, name, age, city, segment, signup date.
Products	60 products	Product ID/name, category, unit cost, unit price, active status.
Sales_Reps	30 representatives	Rep ID/name, region, hire date, team lead, total revenue, rank.
Targets	30 reps × 12 months	Monthly targets from Jan-24 through Dec-24.
Q1_Lookups	8 questions	Lookup functions and linked-sheet retrieval.

Q2_Cleaning	6 questions	Data cleaning and quality diagnostics.
Q3_Aggregation	6 questions	Conditional calculations and business metrics.
Q4_Pivot	5 questions	PivotTable summaries and calculated fields.
Q5_Advanced	5 questions	Ranking, cumulative totals, and multi-condition analysis.

5. Tools and Excel Techniques

- VLOOKUP and INDEX-MATCH: retrieve customer, product, sales representative, team-lead, and monthly target information.
- Data-quality functions and checks: COUNTIF/COUNTIFS, TRIM, case standardisation, MATCH/ISERROR, and duplicate checks.
- Conditional calculations: SUMIFS, COUNTIFS, AVERAGEIFS, IF, and nested IF logic.
- PivotTables: compare revenue by sales region and product category, group dates by month/quarter, and display percentages of totals.
- Advanced analysis: RANK.EQ, LARGE, INDEX-MATCH, TEXTJOIN, and running totals.
- Helper columns: bring category, segment, region, cost, and profit-margin fields into the order-level analysis where required.

6. Project Workflow

1. Inspect the workbook and understand the fields and relationships between worksheets.
2. Connect Orders to Customers, Products, and Sales_Reps using the matching IDs.
3. Run data-quality checks for duplicates, extra spaces, inconsistent case, broken references, inactive products, and unusual discount/status combinations.
4. Use lookup and aggregation formulas to answer the supplied business questions.
5. Build PivotTables or formula-based summaries to compare performance across categories, regions, and time periods.
6. Interpret the results and document limitations before sharing findings with stakeholders.

7. Initial Data Profile

The following figures were calculated from the workbook's order-level data and are descriptive checks, not a complete evaluation of every assessment question.

Metric	Observed value
Order records	2,000
Customers	300
Products	60
Sales representatives	30
Order date range	01 Jan 2024 to 30 Dec 2024
Sum of final_amount across all order records	₹26,572,948.20
Average final_amount per order record	₹13,286.47
Completed orders	1,437 (71.9%)

8. Data Quality Observations

- 20 order ID values appear more than once (each repeated ID counted once).
- 15 customer names contain leading or trailing whitespace.
- 30 order rows reference customer IDs not found in the Customers sheet.
- 0 order rows reference product IDs not found in the Products sheet.
- 0 order rows reference sales-representative IDs not found in the Sales_Reps sheet.
- 174 order rows refer to products marked inactive in the product helper field.
- 58 orders have status Cancelled and a discount greater than 20%. This is a diagnostic pattern, not proof of promo abuse.
- City values include case and whitespace variants; standardisation should be applied in helper columns rather than changing the source data sheet.

Note: These checks are calculated from the supplied workbook data. The workbook's formulas and question cells may depend on Excel recalculation; results can differ if the data or formulas are changed.

9. Business Questions Covered

Q1 – Lookup Functions

Order-to-customer lookup, order-to-product lookup, representative and region lookup, two-way target lookup, discount slabs, chained lookups, and team-lead self-join.

Q2 – Data Cleaning & Quality Diagnostics

Duplicate IDs, whitespace in names, standardised city count, invalid customer references, discontinued products, and cancelled orders with high discounts.

Q3 – Conditional Aggregation

Quarterly revenue by city, high-discount completed orders, average amount for selected category/segment, top-three representative revenue, target-performance tier, and revenue for a selected customer segment and age range.

Q4 – PivotTables

Revenue by region and category, monthly representative pivot, category share of regional revenue, quarterly revenue, and average profit margin by shipping city.

Q5 – Advanced Analysis

Representative ranking, cumulative revenue, second-highest product revenue in the West region, top-five representatives joined into one cell, and highest order amount for a selected city/category.

10. Expected Outcomes

- A repeatable Excel workbook in which answers recalculate when source data changes.
- A clearer view of revenue across regions, categories, and time periods.
- A structured list of data-quality issues for further checking.
- A way to compare sales-representative performance against monthly targets.
- Practical experience with commonly used Excel functions and PivotTables for entry-level data analytics work.

11. Limitations and Recommendations

- A duplicate order ID or unusual discount pattern should be investigated; it does not automatically prove fraud or error.
- Standardise city names and trim text in helper columns to preserve the original data for auditability.
- Review orders with missing customer references and reconcile IDs against the source system.
- Confirm whether revenue totals should include Cancelled, Pending, and Refunded records or only eligible statuses; the initial total above sums final_amount across all rows.
- Open the workbook in a recent version of Excel and recalculate formulas before using question-sheet answers as final submission results.
- Use PivotTables and formulas together to validate important totals independently.

12. Conclusion

The Excel Minor Project – Sales Operations Analytics demonstrates how spreadsheet features can support practical data analysis. By combining lookup functions, quality checks, conditional aggregation, PivotTables, and ranking techniques, the workbook provides a framework for examining sales performance and data reliability. The key lesson is that correct analysis depends not only on formulas, but also on understanding the data, checking its quality, and interpreting results carefully.